

Proteus: Adaptive Scale-Space Analysis for Learning Non-Linear Fuzzy Manifold Memberships (*Draft*)

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**This is a preliminary working version posted for discussion and feedback.
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Abstract

We present Proteus, a two-stage framework for learning explicit, interpretable, and generative models of high-dimensional data supported on non-linear manifolds across multiple nested scales. Proteus separates fast geometric discovery from statistically grounded refinement: Stage 1 performs a scale-space search with an augmented Growing Neural Gas to produce a multiscale scaffold that supports efficient querying and coarse classification. Stage 2 lifts this scaffold to a dynamic simplicial complex and refines it into a piecewise-linear generative core, optionally composing localized invertible warps where curvature exceeds the capacity of a linear approximation. Discrete structural edits (splits, merges/prunes, and warp attachments) are proposed by geometric diagnostics but accepted only when they improve a local free-energy proxy derived from finite-data transition statistics, providing an evidence-based notion of model selection under noise and weak sampling. Finally, a dual-flow reconstruction yields a divergence-consistent probability field on the learned complex, enabling likelihood evaluation, semantic queries under fuzzy constraints, and conditional synthesis within a single representation.

1 Introduction

Motivation, contributions, and roadmap

1.1 Motivation and Gap

High-dimensional data often concentrates near non-linear manifolds whose *structure is scale dependent*: a pattern can be obvious at one granularity and disappear at another, and local intrinsic dimension can vary across the same dataset. In these settings, a useful model should do more than produce an embedding. It should discover characteristic scales with minimal tuning, represent the geometry explicitly enough to be inspected and edited, and expose an inference interface (likelihoods, queries, and conditional generation) without switching representations.

Today this “full stack” is typically assembled from separate tools. Embedding methods and topological diagnostics can reveal shape, but they rarely produce an explicit, generative object. Deep generative models can fit densities, but they often hide geometry and topology behind a parameterization that is difficult to interrogate. Meanwhile, growing-graph and mesh methods can build explicit scaffolds efficiently, but they are frequently governed by heuristic stopping

and ad hoc complexity control, making it hard to tell whether a proposed split or prune reflects real structure or finite-sample noise.

1.2 What Proteus Delivers

Proteus is a two-stage system designed to unify these pieces around a single, inspectable representation. Stage 1 performs fast multiscale discovery and produces a *scaffold*: a hierarchy of coarse graphs that summarize adjacency and local evidence across candidate scales. Stage 2 lifts a selected scaffold to a simplicial complex and refines it into a generative core, while monitoring when a piecewise-linear approximation is insufficient and upgrading capacity only where justified.

Across both stages, Proteus uses a simple control loop: geometry proposes candidate structural edits; a local evidence test decides whether they are accepted; and continuous parameters are settled between accepted edits. The output is an explicit geometric model equipped with a probability field, enabling likelihood evaluation, semantic queries, and conditional synthesis within the same learned object.

1.3 Contributions

- A two-stage architecture that separates scalable multiscale discovery (Stage 1) from evidence-grounded refinement into an explicit simplicial generative core (Stage 2).
- A scale-space controller that discovers characteristic scales and triggers hierarchical decomposition only where supported by routed evidence.
- An evidence-gated mechanism for discrete structure selection (splits, pruning/merges, and optional local non-linear upgrades) using a local free-energy proxy with complexity penalties.
- A torsion-based geometric audit that detects localized failures of piecewise-linear approximation and proposes targeted interventions.
- A dual-flow reconstruction that yields a divergence-consistent probability field on the learned complex for likelihood queries and conditional synthesis.

1.4 Roadmap

Section 2 summarizes Proteus end-to-end and enumerates the learned artifacts. Sections 3 and 4 describe Stage 1 at a single scale and the multiscale controller, respectively. Section 5 introduces simplicial lifting and the core objective; Section 6 presents the torsion audit; Section 7 formalizes evidence-based structure selection; and Section 8 describes the generative and inference interface. Section 9 evaluates model quality and ablations, followed by limitations and scaling and a concluding discussion. Derivations, defaults, and implementation details are provided in the Supplementary Information (SI).

1.5 Related Work

Proteus sits at the intersection of manifold learning, explicit geometric modeling, and probabilistic generative inference. It shares goals with embedding-based manifold learning and topological diagnostics, but targets an explicit object that can be refined and queried rather than an embedding alone. It builds on the tradition of self-organizing and growing-graph methods by pairing fast geometric proposals with an evidence-based acceptance rule for discrete edits. It also complements deep generative modeling by treating flexible invertible maps as localized upgrades to an explicit geometric core, rather than as the sole representation. Finally, its refinement and inference components draw on ideas from discrete geometry (field-like constructions on simplicial complexes) and variational model selection (free-energy proxies) to couple interpretability with a principled notion of structural complexity.

2 Framework Overview

Proteus in one page

2.1 At a Glance

Proteus learns an explicit geometric model of a data distribution together with a principled rule for when to add, remove, or locally upgrade structure. It is organized as a two-stage pipeline:

- **Stage 1 (multiscale discovery):** rapidly explore candidate scales, build coarse distributional scaffolds, and route finite-sample evidence into those scaffolds to determine where further decomposition is warranted.
- **Stage 2 (fixed-scale refinement):** lift a selected scaffold to a simplicial complex at a chosen characteristic scale, refine it into a generative core, and attach additional local capacity only when a geometric diagnostic and an evidence test agree it is justified.

This separation keeps scale discovery cheap and robust, while reserving high-fidelity refinement for the scales and regions supported by data.

2.2 Inputs and Learned Artifacts

Given a dataset of samples, Proteus produces a set of concrete artifacts that together constitute the learned model:

- a **multiscale scaffold hierarchy**: a sequence (and recursion tree) of node–edge graphs with local statistical summaries, used for fast neighborhood queries and for discovering structure across scales;
- a **fixed-scale simplicial complex**: a refined mesh-like representation built from a selected scaffold, serving as an explicit, interpretable generative core;
- a **probability field on the complex**: a divergence-consistent field used for likelihood evaluation, semantic queries, and conditional sampling within the learned geometry;
- optional **localized non-linear upgrades** (“warps”) attached only to patches where curvature exceeds the capacity of a piecewise-linear approximation.

2.3 The Pipeline Contract (Propose, Test, Settle)

Across both stages, Proteus uses the same systems-level learning loop:

- **Propose:** fast geometric diagnostics suggest candidate edits (e.g., add capacity, prune weak structure, or upgrade a patch).
- **Test:** a local evidence score decides acceptance, penalizing unnecessary complexity so that edits triggered by noise or weak sampling are rejected (details in SI).
- **Settle:** between accepted edits, continuous state (geometry and mass/field variables) is re-equilibrated until local statistics stabilize.

This loop is the reason Proteus behaves as a single coherent system rather than a bag of heuristics: geometry suggests where the model is strained, while evidence determines whether a discrete change is warranted.

2.4 What Is Deferred (Progressive Disclosure)

This overview introduces names and roles only. Formal definitions, symbols, thresholds, and derivations are introduced in their owning sections (or the SI) when they become necessary for understanding the mechanism. As a rule, the main text focuses on architecture and logic, while the SI holds derivations, default values, and implementation details.

3 Stage 1: The Distributional Scaffold

Single-scale scaffold construction

Stage 1 is Proteus’ fast “front end.” At a *fixed* scale setting, it constructs a coarse but explicit geometric representation of the data distribution and accumulates the evidence needed to (i) stabilize that representation and (ii) decide whether additional capacity is warranted. The result is a scaffold that is interpretable, queryable, and cheap to update online, and that can later be refined or decomposed by the controller in Section 4.

3.1 What Stage 1 Is For

Stage 1 is designed to do three things efficiently:

1. **Summarize geometry and mass locally:** provide a sparse set of representative nodes with local statistical state.
2. **Route finite-sample evidence:** turn a stream of samples into stable node- and edge-level evidence (hits and transitions) rather than brittle per-sample decisions.
3. **Propose discrete changes:** identify where the current scaffold is under-capacity (suggesting a split) or over-parameterized (suggesting pruning), while deferring acceptance to an evidence gate formalized later (Section 7).

3.2 Scaffold as a Data Structure

The scaffold is a node–edge graph. Each node represents a local region of the data distribution (a proxy for a Voronoi cell around a representative point), and each edge represents adjacency together with accumulated transition evidence induced by local routing. Concretely, Stage 1 maintains:

- **node geometry:** a representative location in data space;
- **node state:** local moment summaries and derived diagnostics (defined below);
- **edge evidence:** statistics that summarize how often local routing connects nearby nodes (used both for topology maintenance and as downstream evidence).

This choice makes the representation explicit (you can inspect nodes and edges) while keeping updates local.

3.3 Node State: What We Track and Why

For each node i , Stage 1 maintains exponentially weighted moving averages (EWMAs) of the first two residual moments, together with derived diagnostics:

- **mean residual $\mathbf{m}_i$:** a coherent drift signal indicating a persistent mismatch between the node’s representative location and the local data it explains.
- **second-moment summary $\mathbf{s}_i$:** used to estimate a local variance proxy.

- **variance proxy** σ_i^2 : a heterogeneity / under-capacity signal. We use the standard moment identity elementwise and summarize it as a scalar trace

$$\sigma_i^2 = \text{tr}(\mathbf{s}_i - \mathbf{m}_i \odot \mathbf{m}_i) = \|\mathbf{s}_i\|_1 - \|\mathbf{m}_i\|_2^2,$$

with $\odot$ denoting elementwise multiplication. Unless stated otherwise, σ_i^2 refers to this scalar summary (e.g., when compared against the split threshold $\tau_{\text{local},i}$).

- **incoherence ratio** ρ_i : a stability-normalized drift score that discounts mean drift when uncertainty is large (a common choice is $\rho_i = \|\mathbf{m}_i\|/(\sigma_i + \varepsilon)$, with $\sigma_i = \sqrt{\sigma_i^2}$; see Appendix: Notation).
- **principal error direction** $\mathbf{u}_i$: an online estimate of the dominant residual direction, used as a placement heuristic when proposing new capacity.

The EWMA’s provide a controlled memory of recent evidence; defaults and derivations (including the chosen EWMA weight schedule) are provided in the SI.

3.4 Per-Sample Routing and Updates

Given an incoming sample, Stage 1 identifies the k nearest nodes and routes evidence locally rather than assigning all credit to a single winner. Neighbors are updated with rank-ordered weights (e.g., a halving schedule $1/2, 1/4, 1/8, \dots$) to achieve robust, localized credit assignment:

1. **select** the k nearest nodes;
2. **update** each selected node’s state via EWMA moment updates using its assigned weight;
3. **accumulate** fractional hit and transition evidence on nodes and edges induced by this routed neighborhood.

This routing step is the source of the “evidence” later used for pruning, for stability checks, and (in Section 4) for deciding which scales and regions deserve additional attention.

3.5 Proposed Discrete Edits (Conceptual)

Stage 1 continuously generates *proposals* for discrete structure changes:

- **Split proposals**: sustained high variance or systematic drift indicates that a single representative is explaining too much heterogeneous mass. A new node is proposed near the stressed node, typically along the estimated principal residual direction $\mathbf{u}_i$.
- **Prune proposals**: nodes or edges with persistently weak routed support are proposed for removal (subject to safety checks that preserve essential connectivity).

Importantly, Stage 1 does not treat proposals as automatically correct. Proposals are *accepted only if* a local evidence test approves them; the acceptance rule and its complexity penalty are formalized in Section 7 (with details in the SI).

3.6 Stabilization at a Fixed Scale

At a fixed scale setting, Stage 1 runs until its local summaries stop drifting in a practical sense. Operationally, we use a simple stabilization criterion based on the equilibration of node-level diagnostics under routed updates (details and default tolerances in the SI). When stabilized, the scaffold is considered “done” at that scale and can be scored and/or decomposed by the controller.

Algorithm 1 Single-scale scaffold update (Stage 1)

```
1: procedure UPDATESCAFFOLDWITHSAMPLE( $\mathbf{x}, \mathcal{G}$ )
2:   Find  $k$  nearest nodes to  $\mathbf{x}$ 
3:   Assign rank-ordered weights to the  $k$  neighbors
4:   for each neighbor  $i$  do
5:     Update EWMA moment state  $(\mathbf{m}_i, \mathbf{s}_i)$  using assigned weight
6:     Update derived diagnostics  $(\sigma_i^2, \rho_i, \mathbf{u}_i)$ 
7:   end for
8:   Accumulate fractional hit/transition evidence on affected nodes/edges
9:   Propose local edits (split/prune) from diagnostics and evidence
10: end procedure
```

3.7 Outputs Handed to the Controller

The output of Stage 1 at a single scale is the stabilized scaffold together with the evidence needed to drive subsequent decisions:

- the node–edge graph geometry (node locations and adjacency),
- per-node state $\{\mathbf{m}_i, \mathbf{s}_i, \sigma_i^2, \rho_i, \mathbf{u}_i\}$,
- routed hit/transition evidence on nodes and edges, and
- a stability status indicating whether local summaries have equilibrated.

Section 4 uses these outputs to search over scales and to recurse only where sufficient evidence exists.

Computationally, per-sample updates are $\mathcal{O}(kd)$, with occasional deferred $\mathcal{O}(d^2)$ work on moves/splits; nearest-neighbor queries are handled with HNSW. Full implementation details are provided in SI S11.

4 Stage 1 Controller: Multiscale Discovery

Scale-space search and recursion

4.1 Why a Scale Controller Is Needed

There is rarely a single “right” resolution at which a dataset’s geometry is both stable and informative. If the representation is too fine, finite-sample noise and local idiosyncrasies trigger spurious structure; if it is too coarse, genuine substructure is collapsed. The role of the Stage 1 controller is to orchestrate repeated single-scale Stage 1 runs, identify characteristic scales, and recurse only when there is enough evidence to justify further decomposition.

4.2 The Scale Knob τ

Stage 1 is governed by a scale parameter τ that acts as a variance/growth threshold: increasing τ makes the scaffold coarser (fewer, broader regions with less pressure to split), while decreasing τ yields finer partitions. Intuitively, τ plays the role of a smoothing scale in classical scale-space analysis (details and calibration are given in SI S2.1).

4.3 Control-to-Threshold Mapping

To keep scale search decoupled from the local adaptation dynamics of Stage 1, we represent scale in a few layers (see SI S2.4):

1. **Control parameter** ($s_{control}$): a normalized knob in $[0, 1)$ used by an optimizer.
2. **Subspace dimensionality** ($D_{subspace}$): the dimensionality of the current sub-problem.
3. **Global threshold** (τ_{global}): a problem-level scale derived from the control parameter,

$$\tau_{global} = -D_{subspace} \log(1 - s_{control}),$$

used to score candidate scales.

4. **Local thresholds** ($\tau_{local,i}$): per-node thresholds that adapt growth to local intrinsic dimension,

$$\tau_{local,i} = -d_{final,i} \log(1 - s_{control}),$$

where $d_{final,i}$ is a smoothed local dimension estimate.

The controller searches using $s_{control}$ (or equivalently τ_{global}), while Stage 1 consumes local thresholds that reflect heterogeneous dimensionality.

4.4 A Scale Response $\Phi(\tau)$

To decide which scales are meaningful, the controller evaluates a scalar response $\Phi(\tau)$ after Stage 1 has stabilized at that scale. Concretely, $\Phi(\tau)$ is a scale-normalized summary of routed evidence: it aggregates per-node fractional hit mass against a local neighborhood volume proxy at scale τ , with the normalization chosen so that responses are comparable across scales (geometric grid ratio and FWHM-based separation given in SI S2.1). Peaks of Φ in $\log \tau$ indicate candidate characteristic scales; broad, flat Φ implies no distinguished scale exists in the searched band.

4.5 Grid Search, Local Refinement, and Selection

We evaluate $\Phi(\tau)$ on a geometric grid $\{\tau_i\}$ to cover orders of magnitude efficiently. The grid ratio is chosen from a scale-space resolution argument (SI S2.1); in practice we use a fixed ratio $r \approx 1/\sqrt{2} \approx 0.71$. At each τ_i , Stage 1 is run to its stabilization criterion and $\Phi(\tau_i)$ is recorded. Candidate maxima are bracketed on the grid and then refined with a lightweight Bayesian optimizer restricted to the bracket to obtain a selected characteristic scale τ^* .

4.6 Recursive Decomposition (Evidence-Limited)

Once a characteristic scale τ^* has been selected for a region, the stabilized scaffold at τ^* induces a partition of the data (via routing to scaffold elements). The controller then recurses on the induced subregions *only* when there is sufficient evidence to support another round of scale discovery. Recursion terminates on any of three conditions: (i) routed evidence in a subregion falls below a minimum-sample floor (SI S9); (ii) the subregion’s scale response $\Phi(\tau)$ is flat on the grid (no significant peak exceeding a small, noise-calibrated margin), indicating no further distinguished scale exists in the searched band; or (iii) a maximum recursion depth is reached. This is what prevents recursion from exploding: without either evidence or a distinguished scale, a region is declared terminal and is handed off (as-is) to downstream refinement.

The meta-level controller is summarized below.

The output of the controller is a hierarchy of stabilized scaffolds together with selected characteristic scales. These outputs provide (i) an interpretable multiscale summary of the data distribution and (ii) warm-starts and scale choices for Stage 2 refinement (Section 5).

Algorithm 2 Recursive scale-space search (Stage 1 controller)

```
1: procedure FINDSCALES( $X$ )
2:   Choose geometric grid  $\mathcal{T} = \{\tau_0 r^j\}_{j=0}^J$  with  $r=1/\sqrt{2}$ 
3:   for each  $\tau \in \mathcal{T}$  do
4:     Run Stage 1 on  $X$  at scale  $\tau$  until stabilized, yielding scaffold  $\mathcal{M}(\tau)$ 
5:     Record response  $\Phi(\tau)$ 
6:   end for
7:    $\mathcal{B} \leftarrow \text{BracketsFromSecondDifference}(\Phi, \mathcal{T})$ 
8:    $\tau^* \leftarrow \text{BayesianOptimize}(\Phi, \mathcal{B})$ 
9:    $\mathcal{M}^* \leftarrow \text{RunStage1ToEquilibrium}(X, \tau^*)$ ;  $\{X_k\} \leftarrow \text{PartitionByMesh}(X, \mathcal{M}^*)$ 
10:  for each partition  $X_k$  do
11:    if  $|X_k| \geq n_{\min}$  then
12:      FINDSCALES( $X_k$ ) ▷ Recurse on sufficiently large sub-regions
13:    end if
14:  end for
15: end procedure
```

5 Stage 2: Simplicial Lifting and the Core Objective

From scaffold to a simplicial generative core

Stage 2 is the refinement stage that turns a stabilized scaffold at a selected characteristic scale τ^* into an explicit *generative* model of the data manifold at that scale. The key upgrade is representational: we lift the Stage 1 node–edge graph into a simplicial complex so that geometry and probability mass can be modeled on higher-dimensional elements (simplices), not only on node-centric Voronoi regions.

5.1 From Scaffold to Complex: Why Lift?

The Stage 1 scaffold is an efficient summary, but it is not yet an ideal domain for generative inference. In particular, likelihood evaluation, sampling, and “field-like” constructions benefit from a representation whose basic units have volume/area (not just points and edges). A simplicial complex provides:

- **local patches:** simplices act as minimal cells that approximate the manifold piecewise-linearly;
- **explicit adjacency:** shared faces encode neighborhood relations needed for local conservation and transport constraints;
- **elementwise modeling:** mass and geometric diagnostics can be attached to cells and propagated through cell adjacency.

This shift is what makes it possible to speak about an explicit generative core and to later reconstruct a probability field directly on the learned geometry (Section 8).

5.2 The Simplicial Representation in Proteus

In Stage 2, the vertices of the complex are the scaffold nodes carried forward from Stage 1. Edges inherit scaffold adjacency, and higher-dimensional simplices are constructed to fill in local neighborhoods, yielding a complex that approximates the manifold at scale τ^* . The complex is dynamic: as refinement progresses, local edits can add or remove vertices and re-triangulate neighborhoods, but the representation remains an explicit simplicial object rather than an implicit parameterization.

The payoff of this representation is that learning and diagnostics can be expressed in terms of simplex-local quantities and their interactions across faces. When the manifold is well approximated piecewise-linearly at the chosen scale, this provides an interpretable generative substrate. When it is not, Stage 2 includes mechanisms that diagnose and locally upgrade capacity (Section 6 and Section 8).

5.3 Warm-Start and Initialization

Stage 2 is run as an intensive refinement pass at fixed τ^* . It warm-starts from Stage 1 by reusing node positions, scaffold connectivity, and accumulated statistics, then initializes an initial simplicial complex from the scaffold graph. The goal is to begin refinement from a plausible geometry rather than from a random mesh. Initialization mechanics and implementation details are provided in SI S9.

5.4 The Learning Objective: Simplex Equilibrium

At a fixed scale τ^* , Stage 2 refines a simplex-associated mass field so that the complex becomes both geometrically and probabilistically consistent with the observed data. We treat simplex masses as latent variables in a generative view of the local transition evidence routed through the mesh (SI S13). **Simplex equilibrium** is the settled configuration of this mass field: a locally optimal tradeoff between data fit, regularity, and model complexity (acceptance criteria and penalties are formalized in Section 7 and detailed in the SI).

Operationally, the equilibrium picture provides a useful separation of regimes. In *plastic* regions, error is reduced by continuous adjustments (local geometry updates and mass reallocation). In *rigid* regions, error accumulates as structural stress indicating that a discrete edit (e.g., adding capacity or upgrading a patch) may be warranted, subject to the evidence gate.

5.5 Node vs. Simplex Equilibrium (Intuition)

Two limiting “balance” conditions help build intuition for why simplex-level modeling produces stable behavior:

- **Node equilibrium:** probability mass is approximately balanced across node-associated regions,

$$P(V_i) \approx 1/N,$$

where V_i denotes the Voronoi-like region explained by node i and N is the number of nodes.

- **Simplex equilibrium:** probability mass is approximately balanced across simplices,

$$P(C_j) \approx 1/M,$$

where C_j denotes simplex j and M is the number of simplices.

These are not identical: simplex masses resolve mass distribution at the level of local patches rather than node cells. The correspondence between them has two parts. (i) *Static correspondence* (proved in SI S12.2 as a finite-sample theorem): the deterministic decomposition $P(V_i) = \sum_{C \ni i} \alpha_{i,C} P(C)$ combined with multinomial concentration shows that empirical simplex equilibrium with tolerance ϵ_{sim} implies node equilibrium up to (a) an irreducible structural floor controlled by simplex shape quality and star regularity, (b) a residual equilibration term proportional to ϵ_{sim} , and (c) a sampling term of order $\sqrt{\log(M/\delta)/H}$. (ii) *Dynamic equilibration* (informal mechanism, SI S12.2): the variance-cap split rule and the local evidence test (Section 7) drive simplex masses toward $1/M$ on the manifold interior; the loop stalls only at structural junctions, where the irreducible floor identifies the dimensionality-junction signature. A formal

Lyapunov argument and a quantitative stuck-junction lemma are open theoretical questions catalogued in SI S15.

The remaining Stage 2 sections build on this setup: Section 6 introduces a geometric audit for when the piecewise-linear approximation fails, and Section 8 describes how the settled complex supports likelihoods, queries, and sampling.

6 Stage 2 Diagnostics: Torsion Audit

When piecewise-linear geometry fails

6.1 Why Piecewise-Linear Geometry Fails

Stage 2 models the manifold with a simplicial complex, which is necessarily a piecewise-linear (PL) approximation. This approximation is often excellent at a chosen scale τ^* , but it has an unavoidable failure mode: sufficiently curved structure cannot be made globally consistent by local vertex moves and re-triangulation alone. In Proteus, this failure is detected through the behavior of the local residual mean field. If the PL model is locally adequate, residual means behave (approximately) like a conservative correction field; when curvature, torsion-like twisting, or mismatched dimensionality forces non-conservative behavior, the residual field develops rotational components that continuous refinement cannot remove.

6.2 Discrete Torsion on a Simplex

Proteus formalizes this diagnostic with a discrete 2-form that measures how far the residual mean field deviates from being locally conservative on each simplex. For a d -simplex $S = \{v_0, \dots, v_d\}$ with vertex positions $\{\mathbf{w}_{v_i}\}$ and residual means $\{\mathbf{m}_{v_i}\}$, define barycenters $\bar{\mathbf{w}}$ and $\bar{\mathbf{m}}$ and construct edge and residual matrices (relative to the barycenter):

$$E = [\mathbf{w}_{v_1} - \bar{\mathbf{w}} \ \cdots \ \mathbf{w}_{v_d} - \bar{\mathbf{w}}], \quad M = [\mathbf{m}_{v_1} - \bar{\mathbf{m}} \ \cdots \ \mathbf{m}_{v_d} - \bar{\mathbf{m}}].$$

We drop v_0 in the columns of E and M to remove the linear redundancy induced by centering over all $d+1$ vertices (equivalently, any one vertex can be dropped; Ω_S below is invariant to the choice). The simplex torsion 2-form is the antisymmetric part of the local geometry-to-residual quadratic form,

$$\Omega_S = M^\top E - E^\top M.$$

If the residual mean field were (locally) the gradient of a potential, the implied Jacobian $J \approx ME^+$ would be symmetric and $\Omega_S = E^\top (J - J^\top) E$ would vanish; in particular Ω_S scales quadratically with simplex size, which is why the ratio $R_S = \kappa_S / \tau^*$ below is dimensionless. Intuitively, Ω_S is a discrete analogue of curl: it is (approximately) zero when residual means are consistent with a potential-like correction over the simplex, and it grows when the residual field “swirls” around the simplex in a way that signals PL mismatch. We summarize torsional stress with a scalar

$$\kappa_S = \|\Omega_S\|_F.$$

6.3 The Torsion Ladder (Triage Policy)

Torsion is a *proposal signal*. It does not decide whether an intervention is warranted; it only localizes where the PL approximation is strained and suggests appropriate classes of corrective actions. We normalize torsional stress by the fixed operating scale and define a torsion ratio

$$R_S = \kappa_S / \tau^*,$$

Placeholder schematic: Torsion Ladder triage with thresholds on $R_S = \kappa_S/\tau^*$ and actions (keep, split, mini-NSF).

Figure 1: Torsion Ladder triage with thresholds on $R_S = \kappa_S/\tau^*$ and resulting actions (keep, split, mini-NSF).

which compares local non-conservativity to the characteristic variance scale at which Stage 2 is operating.

The **Torsion Ladder** is a simple triage policy over R_S :

- **Ignore / monitor:** very small ratios are treated as sampling noise or numerical effects; the simplex is monitored but no action is taken.
- **Refine geometry:** when curvature appears tolerable, Stage 2 focuses on additional local settling (continuous refinement and mass re-equilibration) without changing model class.
- **Add discrete capacity:** persistent intermediate ratios propose a localized split that adds vertices and re-triangulates the stressed patch. Split placement is aligned with the local torsion signal to target the directions in which the PL approximation is failing.
- **Escalate to non-linear upgrades:** when curvature is too strong for a stable PL patch, the ladder proposes attaching a localized invertible warp to the patch (details in Section 8 and SI S7).

All ladder actions beyond monitoring are evaluated by the same evidence-based acceptance rule used throughout Proteus (Section 7); torsion determines *where* and *what kind* of intervention to try, while evidence determines whether it should be kept. The numerical thresholds that separate the ladder’s regimes (given in SI S4) are empirical defaults calibrated on synthetic benchmarks; they are intentionally coarse and may require per-dataset adjustment, but because acceptance is always gated by the evidence test, the ladder’s role is only to propose an appropriate *class* of action.

6.4 Shape-Quality Guard and Safe Edits

Discrete edits must also preserve numerical stability. In particular, aggressive splits can create sliver simplices that destabilize downstream updates. Proteus therefore screens split proposals with a simplex quality score

$$Q_S = dr_{\text{in}}/R_{\text{circ}} \in (0, 1],$$

where r_{in} and R_{circ} are the inradius and circumradius of the simplex. If a proposed split would produce poor-quality children, it is rejected in favor of a more conservative insertion strategy (e.g., a centroid-based “star split”), with defaults and variants specified in the SI.

Finally, Stage 2 pruning is arbitrated at the simplex level: deletions require agreement across the simplex star of a node, with geometric veto rules to protect structurally critical “tent-pole” vertices (SI S11). This keeps the refinement dynamics from removing vertices that are topologically or geometrically essential.

7 Evidence-Based Structure Selection

Geometry proposes; evidence accepts

7.1 Why a Single Disposer Is Needed

Proteus proposes discrete edits from several sources: Stage 1 proposes splits and pruning from scaffold diagnostics (Section 3); Stage 2 proposes targeted interventions from torsion and element-quality audits (Section 6); and optional non-linear upgrades can be proposed when local curvature cannot be resolved by piecewise-linear refinement (Section 8). Without a shared acceptance rule, these mechanisms would behave like independent heuristics, each with its own implicit notion of “improvement.” Instead, Proteus uses a single *local evidence criterion* to arbitrate *all* discrete edits. This keeps structure selection evidence-limited under finite data and makes proposals comparable across edit types.

7.2 Defining the Affected Region $\mathcal{R}$

Discrete edits are local: they change only a small portion of the scaffold/complex (e.g., splitting a vertex neighborhood, removing a weak node, or attaching a patch-wise warp). We therefore score edits on a local region $\mathcal{R}$ that contains:

1. all nodes whose simplex stars would change under the proposed edit, and
2. a small additional neighbor ring to stabilize normalizers in the transition router used for likelihood evaluation.

The goal of $\mathcal{R}$ is to include every term that materially changes under the edit, while keeping scoring local enough to be efficient and to avoid re-optimizing the full model for each proposal (formal details in SI S3.4).

7.3 A Local Free-Energy Proxy

Let $\mathcal{L}_{\mathcal{R}}$ denote the transition log-likelihood restricted to sources in $\mathcal{R}$, evaluated under the geometry-induced router $q(j \mid i; m)$ (SI S13). Let $\log p(m \mid M)$ be a prior on the simplex mass field under candidate topology M (e.g., a Dirichlet composition prior or a smoothness prior such as a logistic-GMRF). For a candidate edit producing a local topology choice M (keep vs. split vs. merge/prune vs. warp attachment), we warm-start the local mass field from the current solution and settle it to a local MAP configuration $\hat{m}$ while keeping the exterior fixed. The candidate is then scored by a free-energy proxy

$$F(\mathcal{R}; M) = -\mathcal{L}_{\mathcal{R}}(\hat{m}; M) - \log p(\hat{m} \mid M) + \frac{1}{2} k_M \log N_{\text{eff}, \mathcal{R}},$$

where k_M is the number of free parameters introduced by topology M within $\mathcal{R}$, and $N_{\text{eff}, \mathcal{R}}$ is the effective transition count supporting the local score. The first term rewards topologies that better explain observed transitions, the second term regularizes the mass field, and the final term penalizes unnecessary complexity using a BIC-like form. This proxy provides an operational model-selection rule under finite data: add capacity only when it yields enough evidence-weighted improvement to overcome its complexity cost (derivation and router specifics in SI S3.4 and SI S13). We note that the BIC form assumes standard regularity conditions that can be violated at the boundary of the Dirichlet–multinomial (e.g., when some $n_{i \rightarrow j} = 0$); we use BIC as a computationally cheap default and treat WAIC/PSIS-LOO as asymptotically more general alternatives when needed (SI S3.4).

7.4 Acceptance Protocol

The accept/reject loop follows the global propose–test–settle pattern introduced in Section 2:

1. **Propose** an edit from a geometric or statistical diagnostic (split/prune/merge/warp).

2. **Localize** the affected region $\mathcal{R}$ induced by the edit.
3. **Warm-start and settle** continuous variables locally (mass field and any local geometry parameters needed for scoring), holding the exterior fixed.
4. **Score** the candidate with $F(\mathcal{R}; M)$ and compare to the keep-topology baseline.
5. **Accept** only if free energy decreases by a margin; otherwise reject and continue settling under the current topology.

The margin prevents churn from noisy or marginal improvements; default values and variants are provided in the SI.

7.5 Avoiding Double Counting with Reconstruction

After topology selection and settling, Proteus reconstructs a probability field on the accepted complex (Section 8). This reconstruction is *conditioned* on the accepted topology and its settled mass/geometry. It is not treated as an additional independent likelihood term for the same transition evidence already used in $\mathcal{L}_{\mathcal{R}}$. This separation avoids double counting: the evidence gate selects structure using local transition statistics, and the reconstruction step converts the accepted, settled representation into an inference-ready probability field.

8 Generative Modeling and Inference Interface

Likelihoods, queries, and conditional synthesis

8.1 What Is Reconstructed

After structure selection and settling, Proteus produces an inference-ready probability field on the learned complex. The core idea is to reconstruct a conservative face-wise “pressure” / flux field $\{p_f\}$ on the complex that (i) matches the empirical evidence accumulated during training and (ii) satisfies discrete conservation constraints implied by a valid probability field. This reconstruction is performed *after* topology selection and is conditioned on the accepted complex and its settled mass field $\{m_S\}$ and geometry (Section 7). Conceptually, $\{m_S\}$ fixes *how much* mass sits in each simplex, while $\{p_f\}$ specifies the *interpolated flux geometry* on faces consistent with that mass under discrete conservation. The two are therefore *complementary*, not redundant: $\{m_S\}$ is used for structure selection via the transition likelihood (and is frozen once accepted); $\{p_f\}$ is the inference-time interpolant used for density evaluation, queries, and sampling, and is not reused as a selection signal for the same evidence.

8.2 Dual Flow Reconstruction (Minimal View)

Proteus accumulates empirical flux evidence on simplex faces using a **dual flow**. For a sample $\mathbf{x}$ assigned to a winning simplex S , we project the residual onto outward face normals and distribute a nonnegative pressure increment to each facet $f \in \partial S$:

$$\Delta p_f \propto \max(0, (\mathbf{x} - \bar{\mathbf{w}}_S)^\top \mathbf{n}_f); \quad p_f \leftarrow p_f + \Delta p_f.$$

Aggregating these updates across samples yields empirical tallies $\{\hat{p}_f\}$ on faces. We then solve for face pressures $\{p_f\}$ on the dual graph $\mathcal{G}_{\text{dual}}$ (nodes are faces; edges connect adjacent faces) by balancing fidelity to evidence with discrete conservation constraints. This yields a quadratic objective

$$\min_{\{p_f\}} \sum_f \lambda (p_f - \hat{p}_f)^2 + \sum_S \mu \|A_S \mathbf{p}_S\|_2^2,$$

where $\mathbf{p}_S$ collects the pressures on the faces incident to simplex S and A_S is a local stencil that penalizes violations of conservation within S . Solving this objective produces a conservative face-pressure field that induces a piecewise-linear probability gradient field on the complex. Discretization choices and solver variants are provided in SI S4.

8.3 Querying the Learned Model

A point’s association with the learned geometry is intrinsically *fuzzy* rather than hard-categorical. Each point $\mathbf{x}$ is represented by barycentric coordinates $\{\beta_v(\mathbf{x})\}_{v \in V(S)}$ in its containing (or nearest) simplex S , with $\beta_v \geq 0$ and $\sum_v \beta_v = 1$; these act as soft vertex memberships. At the simplex level, membership in a user-defined region (a subset of simplices, a density band, a topological component, or a label class) is expressed as a probability over the restricted simplex set, normalized by mass. This is the sense in which Proteus learns *fuzzy manifold memberships*: both vertex- and region-level associations are real-valued and smoothly varying, not crisp assignments.

Given a query point $\mathbf{x}$, Proteus then supports common inference-time operations in a single representation:

- **Density / likelihood evaluation:** locate the containing simplex (or nearest via barycentric projection), evaluate the reconstructed field locally, and combine with simplex mass to obtain a density estimate (details in SI S13).
- **Anomaly scoring:** compute a low-density score under the learned field, optionally normalized by local scale; anomalies correspond to points that cannot be assigned high likelihood in any nearby simplex.
- **Fuzzy membership and semantic queries:** given a constraint $\mathcal{C}$ defining a subset of simplices (label, density band, topological component), report the membership score $\mu_{\mathcal{C}}(\mathbf{x}) = \sum_{S \in \mathcal{C}} \Pr(S | \mathbf{x})$ where $\Pr(S | \mathbf{x})$ is the posterior simplex membership under the learned field; fuzzy set operations (intersection, union, complement) combine by standard probabilistic or fuzzy-logic rules on the simplex sets.

Boundary faces are handled with a no-exterior-flux (Neumann-like) convention as in the SI.

8.4 Sampling and Conditional Synthesis

Sampling proceeds by drawing a simplex proportional to its settled mass, drawing barycentric coordinates within that simplex, and mapping back to data space. Conditional synthesis follows the same pipeline but conditions the simplex selection step on constraints, for example by:

- selecting only simplices whose cached attributes satisfy the constraint,
- reweighting simplex masses by a constraint likelihood factor, or
- performing local constrained search on the simplex adjacency graph when constraints are relational or path-based.

The key design choice is that conditioning occurs at the level of the explicit complex: constraints prune or reweight a finite set of simplex candidates before sampling, enabling interpretable conditional generation without changing model class.

8.5 Constraint Extraction

Because Proteus routes evidence through explicit scaffold/complex elements, it can summarize structure from a selected subset of data or a query-defined region. Given a subset (or a soft selection) of points, we route them through the learned complex and aggregate:

Algorithm 3 Inference-time interface (high level)

```
1: procedure QUERYDENSITY( $\mathbf{x}$ )
2:   Locate containing simplex  $S$  (or nearest via barycentric projection)
3:   Evaluate local density from simplex mass and reconstructed field on  $S$ 
4:   return  $\log p(\mathbf{x})$ 
5: end procedure
6: procedure SAMPLE
7:   Sample simplex  $S$  with probability proportional to its mass
8:   Sample barycentric coordinates in  $S$  and map to data space
9:   Apply local warp if attached to the sampled patch
10:  return sample  $\mathbf{x}$ 
11: end procedure
12: procedure CONDITIONALSAMPLE(constraints)
13:   Select/reweight a candidate simplex set consistent with constraints
14:   Sample simplex and barycentric coordinates as in SAMPLE
15:   return conditional sample  $\mathbf{x}$ 
16: end procedure
```

- **mass and density summaries** over the visited simplices,
- **transition summaries** induced by routed neighbor assignments, and
- **geometric summaries** such as dominant directions, local intrinsic dimension estimates, or torsion hot spots.

These aggregates can be exported as constraints or predicates for downstream systems (e.g., “generate samples consistent with this subregion”) while keeping the explanation grounded in explicit simplices and their adjacency.

8.6 Optional Warps: Global vs. Patch-Wise

When torsion indicates that curvature exceeds what a stable PL patch can represent at scale τ^* , Proteus can propose localized non-linear upgrades. We summarize curvature prevalence with a torsion coverage statistic P_κ (the fraction of simplices whose torsion ratio exceeds a curvature threshold). If curvature is widespread (high P_κ), a shallow global invertible map can provide broad non-linear capacity; if curvature is localized (low P_κ), Proteus instead attaches compact patch-wise warps (e.g., small spline flows) only to high-torsion regions identified on a facet-adjacency graph. In both cases, the core representation remains the simplicial complex: the probability field is reconstructed in the warped coordinates, and the model in the original space is obtained by pullback through the learned invertible map. Training budgets, patch identification heuristics, and defaults are given in SI S7.

For evaluation, we report generation metrics (MMD) and log-likelihood as detailed in Section 9 and SI S5.

9 Evaluation

Model quality and ablations

We evaluate Proteus as an end-to-end geometric generative representation: does it recover the right qualitative structure, does the learned complex remain geometrically faithful, and does the resulting probability field support likelihood queries and sampling? Our evaluation is therefore model-centric rather than task-centric. Concretely, we measure (i) topological faithfulness of

the learned complex, (ii) geometric reconstruction quality, (iii) held-out likelihood under the reconstructed field, and (iv) sample quality. Each metric is chosen to correspond to a specific owned object in the architecture: simplicial lifting and refinement (Section 5), torsion-driven diagnostics (Section 6), evidence-based structure selection (Section 7), and the dual-flow inference interface (Section 8).

9.1 Experimental Setup

9.1.1 Datasets

We use synthetic datasets with known ground-truth structure (e.g., nested spheres of varying intrinsic dimension, linked tori, variable-density manifolds, and dimensionality junctions) to enable quantitative topology comparisons. We also evaluate on real-world benchmark datasets to assess robustness when ground truth is unavailable. Dataset details and preprocessing choices are provided in SI S5.

9.1.2 Protocol

Unless otherwise stated, we train on an 80/20 split with multiple random seeds and report mean $\pm$ std. For each dataset, Stage 1 discovers characteristic scales and triggers recursive decomposition only when evidence thresholds are met (Section 4). Stage 2 refines the selected representation at the chosen scale, applies torsion-driven proposals as needed (Section 6), and uses the free-energy disposer for all discrete edits (Section 7) before reconstructing the probability field (Section 8). Full protocol, defaults, and evaluation settings are in SI S5 and SI S10.

9.2 Metrics

Our metrics directly assess the learned representation:

- **Topological accuracy (PH):** persistent homology distances between the learned complex and ground-truth synthetic manifolds, evaluating whether lifting and refinement recover correct topological signatures (Section 5–Section 6).
- **Reconstruction error:** average distance from data points to their projection on the learned complex, measuring geometric fidelity of the piecewise-linear (and optionally warped) representation (Section 5 and Section 6).
- **Held-out log-likelihood:** average log-likelihood under the reconstructed probability field, reflecting both structure selection (Section 7) and the dual-flow reconstruction (Section 8).
- **Generative quality (MMD):** distributional similarity between generated samples and held-out data, measuring whether the inference interface supports faithful synthesis (Section 8).

Metric definitions, hyperparameters, and evaluation conventions are provided in SI S5 and SI S10.

9.3 Baselines

We compare Proteus to representative baselines that cover the common “toolchain” alternatives: embedding-plus-density pipelines (e.g., UMAP with a downstream density estimator), and end-to-end generative models (e.g., VAEs). Where applicable, we also include topology- or geometry-focused baselines that produce explicit structures but lack an integrated likelihood/sampling interface. Baseline configurations are detailed in SI S5.

Table 1: Placeholder results: topological accuracy (PH distance), reconstruction error, log-likelihood, and MMD (lower is better where applicable).

Method	PH Dist.	Recon Err	Log-Lik	MMD
Proteus (full)	—	—	—	—
No scale controller	—	—	—	—
No Stage 1 warm-start	—	—	—	—
No torsion audit	—	—	—	—
No evidence disposer	—	—	—	—
No dual flow	—	—	—	—
Baseline (UMAP+KDE)	—	—	—	—
VAE	—	—	—	—

Placeholder ablation plots: effects of removing the scale controller, torsion audit, evidence disposer, or dual flow on PH distance, reconstruction error, and MMD across datasets.

Figure 2: Ablations (placeholders): effects of removing the scale controller, torsion audit, evidence disposer, or dual flow on PH distance, reconstruction error, and MMD across datasets.

9.4 Ablation Studies

We conduct ablations that correspond directly to owned objects in the architecture:

- **No scale controller:** replace multiscale discovery (Section 4) with a fixed, single-scale run to measure the importance of characteristic-scale selection and evidence-limited recursion.
- **No Stage 1 warm-start:** initialize Stage 2 without the stabilized scaffold (Section 3) to quantify the benefit of coarse-to-fine refinement.
- **No torsion audit:** disable the torsion ladder (Section 6) to test whether curvature-driven failures can be handled by geometry-only refinement and variance-based splits.
- **No evidence disposer:** accept discrete edits using heuristic triggers rather than the free-energy proxy (Section 7) to measure the role of evidence-based structure selection under finite data.
- **No dual flow:** disable probability-field reconstruction (Section 8) to isolate the contribution of the inference interface to likelihoods and generation.

10 Limitations and Scaling

Where the method can break; how it scales

10.1 Limitations

Proteus is designed to be evidence-limited and robust, but there are practical regimes where performance or stability can degrade:

- **Very high ambient dimension:** without an initial projection or feature compression, nearest-neighbor search and per-sample updates become expensive and local moment estimates become noisier. This primarily affects Stage 1 stabilization and scale discovery (Section 3 and Section 4).

- **Strong anisotropy and overlapping scales:** highly anisotropic densities can blur scale separation, requiring denser scale evaluation and/or longer stabilization at candidate scales to reliably detect characteristic maxima (Section 4).
- **Sparse local support:** in regimes with weak sampling at high intrinsic dimension, simplex-local diagnostics (including torsion) are noisier. Evidence thresholds and quality guards mitigate instability, but can lead to conservative (underfit) representations unless stronger priors or more data are available (Section 6 and Section 7).
- **Widespread nonlinearity:** when curvature is broad and highly non-stationary, patch-wise upgrades may be insufficient without increased global capacity or additional inductive bias (Section 8).

10.2 Typical Operating Envelopes (Indicative)

These ranges provide a coarse sense of feasibility rather than guarantees:

- $N \in [10^5, 10^7]$, $d \in [64, 2048]$ after optional projection; per-level wall-time dominated by HNSW queries and moment updates.
- HNSW: $M \in [16, 32]$, $\text{efConstruction} \in [100, 400]$, $\text{efSearch} \in [50, 200]$; increase efSearch near convergence for stability.
- Memory: nodes+links+simplexes typically within $\sim 10\%$ growth per recursion level due to Ladder and prune rules.

10.3 Complexity and Bottlenecks (Tied to Owned Objects)

The dominant costs can be understood in terms of the main architectural objects:

- **Stage 1 updates (Section 3):** nearest-neighbor queries plus k local EWMA updates dominate. Per-sample compute is $\mathcal{O}(kd)$ aside from ANN search, with occasional higher-order work on deferred moves/splits.
- **Scale controller (Section 4):** controller cost is Stage 1 cost multiplied by the number of evaluated grid points and (evidence-limited) recursion levels. In practice, most partitions terminate quickly once evidence thresholds are not met.
- **Torsion audit (Section 6):** torsion and simplex-quality computations are more expensive per element but are applied selectively (on flagged patches or topology events) rather than on every simplex at every step.
- **Dual flow reconstruction (Section 8):** reconstruction solves a quadratic objective on the dual graph of faces; it is typically near-linear per solver pass in the number of faces and is performed after settling.

Asymptotic envelopes, solver notes, and implementation details are provided in the SI.

10.4 Scaling Summary

Overall memory grows with the number of scaffold nodes, edges, and simplices. Evidence gating and audit-driven proposals are designed to prevent uncontrolled growth: additional structure must be locally justified before it is retained. Wall time is typically dominated by ANN queries and moment updates in Stage 1, with Stage 2 costs concentrated in localized audits and reconstruction steps.

11 Conclusion

Summary and outlook

Proteus targets a common practical reality in manifold learning: practitioners often stitch together embeddings, topological diagnostics, density estimators, and geometric heuristics, yet still lack a single representation that is simultaneously explicit, generative, and governed by a principled rule for discrete structure selection. Proteus addresses this by combining fast multiscale discovery with evidence-grounded refinement into an explicit simplicial generative core.

At a high level, the contribution is a coherent stack of owned objects:

- a **distributional scaffold** that routes finite-sample evidence into an explicit node–edge representation (Section 3),
- an **evidence-limited scale controller** that selects characteristic scales and recurses only where supported (Section 4),
- a **simplicial lift and refinement objective** that turns the selected scaffold into a generative core (Section 5),
- a **torsion audit** that localizes where piecewise-linear approximation fails and proposes targeted interventions (Section 6),
- a **single evidence disposer** that arbitrates all discrete edits under finite data (Section 7), and
- a **dual-flow inference interface** that reconstructs a probability field for likelihoods, queries, and conditional synthesis (Section 8).

The evaluation (Section 9) supports the claim that each piece is necessary: removing scale discovery, torsion diagnostics, evidence-based selection, or reconstruction degrades topological faithfulness, geometric fidelity, or generative performance.

Proteus also aims to reduce dependence on opaque tuning by grounding key decisions in stability criteria, routed evidence, and complexity penalties, with derivations and defaults provided in the SI. This makes the learned model easier to inspect, debug, and adapt across datasets.

Future work. Immediate directions suggested by this architecture include (i) streaming and continual-update variants of Stage 1 and Stage 2 that preserve evidence-based edit arbitration under non-stationarity, (ii) tighter theory connecting equilibrium objectives, torsion diagnostics, and finite-sample selection guarantees, and (iii) richer but still localized non-linear upgrades that preserve the interpretability and queryability of the explicit complex.

12 References Prep

Draft list to formalize

This section collects potential references to be formalized and cited.

Manifold Learning: Isomap (Tenenbaum et al., 2000); Locally Linear Embedding (Roweis & Saul, 2000); t-SNE (van der Maaten & Hinton, 2008); UMAP (McInnes, Healy, Melville, 2018).

Hierarchical and Density Clustering: HDBSCAN (Campello, Moulavi, Sander, 2013/2015).

Scale-Space Theory: Lindeberg, Scale-Space Theory (1994, 1998).

Growing Neural Gas and Variants: GNG (Fritzke, 1995) and node-centric extensions.

Discrete Geometry and Topology: Discrete exterior calculus / discrete differential forms (Desbrun et al.); Divergence Theorem references (standard vector calculus texts); Persistent homology and Betti numbers (Edelsbrunner & Harer).

Generative Models: VAEs (Kingma & Welling, 2013/2014); GANs (Goodfellow et al., 2014); Normalizing Flows (Dinh et al., 2014/2016; Papamakarios et al., 2021).

Appendix: Notation

We summarize the core symbols used throughout the paper.

- $m_i \in \mathbb{R}^d$: EWMA residual mean at node i .
- $s_i \in \mathbb{R}^d$: EWMA of squared residual at node i (elementwise).
- $\sigma_i^2 \in \mathbb{R}$: scalar variance proxy $\sigma_i^2 = \text{tr}(s_i - m_i \odot m_i) = \|s_i\|_1 - \|m_i\|_2^2$; $\sigma_i = \sqrt{\sigma_i^2}$.
- $\rho_i = \|m_i\|_2 / (\sigma_i + \varepsilon)$: incoherence ratio at node i .
- k : neighborhood size (Stage 1: $k=8$; Stage 2: $k \approx d$).
- D_{subspace} : current working dimensionality; $d_{\text{final},i}$: smoothed local intrinsic dimension at node i .
- $s_{\text{control}} \in [0, 1)$: optimizer control; $\tau_{\text{global}} = -D_{\text{subspace}} \log(1 - s_{\text{control}})$.
- $\tau_{\text{local},i} = -d_{\text{final},i} \log(1 - s_{\text{control}})$: per-node growth threshold.
- $\Phi(\tau)$: scale-space response evaluated at threshold τ .
- Ω_S : simplex 2-form (discrete torsion) on simplex S ; $\kappa_S = \|\Omega_S\|_F$.
- $R_S = \kappa_S / \tau^*$: torsion ratio used by the Torsion Ladder at the fixed scale τ^* .
- $\tilde{\rho}$: neighbor-normalized incoherence; $\text{CV} = \text{std}(\tilde{\rho}) / \text{mean}(\tilde{\rho})$.